

CSE 4849

Human Computer Interaction (HCI)

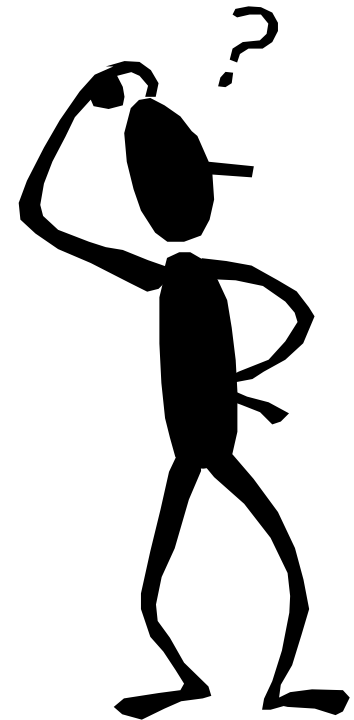
Lecture 6 – The process of interaction design

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Hasan Mahmud | hasan@iut-dhaka.edu

Overview

- What is involved in Interaction Design?
 - Importance of involving users
 - Degrees of user involvement
 - What is a user-centered approach?
 - Four basic activities of Interaction design
- Some practical issues
 - Who are the users?
 - What are 'needs'?
 - Where do alternatives come from?
 - How do you choose among alternatives?
- A simple lifecycle model for Interaction Design
- Lifecycle models from software engineering
- Lifecycle models from HCI



What is involved in Interaction Design?

- It is a process:
 - A goal-directed problem solving activity informed by intended use, target domain, materials, cost, and feasibility
 - A creative activity
 - A decision-making activity to balance trade-offs, about balancing conflicting requirement
- It is a representation:
 - A plan for development
 - A set of alternatives and successive elaborations

Importance of involving users

- Expectation management
 - Make sure that the users' views and expectation of the new product are realistic
 - Ensure that, there are no surprises, no disappointments for users when the product arrives
 - Timely training ensured to manage expectations
 - Communication, but no hype
- Ownership
 - Make the users active stakeholders
 - More likely to forgive or accept problems
 - Can make a big difference to acceptance and success of product

Degrees of user involvement

- Member of the design team
 - Full time: constant input, but lose touch with users if project takes longer time
 - Part time: patchy input, and very stressful
 - Short term: inconsistent across project life
- Newsletters and other dissemination devices
 - Reach wider selection of users
 - Need communication both ways
- Combination of these approaches

What is a user-centered approach?

User-centered approach is based on:

- **Early focus on users and tasks:** directly studying cognitive, behavioural, anthropomorphic and attitudinal characteristics
- **Empirical measurement:** users' reactions and performance to scenarios, manuals, simulations and prototypes are observed, recorded and analysed
- **Iterative design:** when problems are found in user testing, fix them and carry out more tests

Four basic activities of interaction design

There are four basic activities in Interaction Design:

1. Identifying needs and establishing requirements

- know who our target users are
- what kind of support an interactive product could usefully provide.

2. Developing alternative designs

- Suggesting ideas for meeting the requirements in the form of alternative designs
- Contains two sub-activities: Conceptual design and physical design
- Conceptual design involves producing the conceptual model for the product, and a conceptual model describes what the product should do, behave and look like.
- Physical design considers the detail of the product including the colors, sounds, and images to use, menu design, and icon design.

3. Building interactive versions of the designs

- This requires an interactive version of the designs to be built, but that does not mean that a software version is required.

4. Evaluating designs

- Evaluation is the process of determining the usability and acceptability of the product or design that is measured in terms of a variety of criteria including the number of errors users make using it, how appealing it is, how well it matches the requirements, and so on.

Identifying needs and establishing requirements

- Users vs Stakeholders
 - Users: people who will directly interact with the design.
 - Stakeholders: people who will be affected by the design.
- What do the users need to do?
 - Although it is tempting for a designer to design what they would want but a good designer is able to design for the needs of the users of a product.
- Different types of requirements:
 - Functional: what should the product do?
 - Data: what types of data should the product handle?
 - Environmental: physical/social/organisational/technical
 - User: characteristics of the target user group
 - Usability: important usability goals and measures

Identifying needs and establishing requirements

- Techniques for Gathering Data
 - Questionnaires: useful for asking specific questions from a number group of people, questionnaires are often combined with other techniques
 - Interviews: useful for exploring issues that arise as the result of asking questions, but they are time-consuming
 - Focus Groups: good for gaining a consensus view of an issue and/or highlighting areas of conflict/disagreement
 - Observations: shadowing people as they work with a system can provide important insight into what people actually do (compared to what they say they do)
 - Studying Documentation: existing documentation can be a good source of information about user activities

Identifying needs and establishing requirements

- Scenario
 - An informal narrative description of a user performing a task that allows the exploration of the context, needs and requirements
- Use Case
 - A description of how a user (actor) would interact with a system to achieve a task, allowing the description of the interactions required throughout

Developing Alternative Designs

- Conceptual Design
 - Conceptual models for products that describe what a product should do and how the product should behave.
- Detailed Design
 - Detailed models for products that describe aspects of an interface, e.g. colours, sounds, images, icons, menus, etc.

Developing Alternative Designs

- Generating design ideas
 - Methods for generating design ideas
- Cross-fertilisation of existing ideas
 - e.g. analogy, bisociation, case-based reasoning, mutation
- Evolution of existing ideas
 - e.g. addition of new features, extrapolation of trends
- Reproduction of existing ideas
 - e.g. application of ideas to new domains

Building Interactive Versions of Designs

- Evaluating interactive products
 - Users need to evaluate interactive versions of designs
- Interactive version $\neq$ full implementation
 - e.g. paper-based prototypes, role-playing users
- Prototyping products for users
 - Reduces confusion between designers and users

Evaluating Designs

- Observation of users
 - e.g. number of errors made using the design
- Questionnaires of users
 - e.g. how appealing is the design
- Objective evaluation
 - e.g. how well does the design match requirements

Some practical issues

- Who are the users?
- What are 'needs'?
- Where do alternatives come from?
- How do you choose among alternatives?

Who are the users/stakeholders?

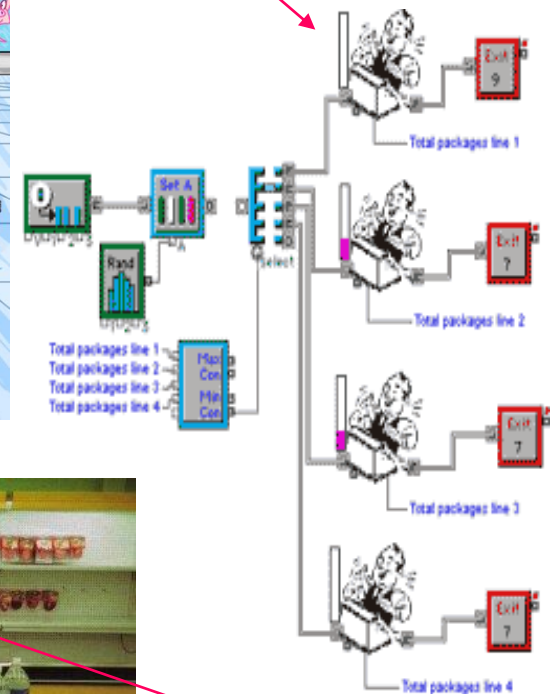
- *"people or organizations who will be affected by the system and who have a direct or indirect influence on the system requirements"* (Kotonya and Sommerville, 1998)
- Not as obvious as you think:
 - those who interact directly with the product
 - those who manage direct users
 - those who receive output from the product
 - those who make the purchasing decision
 - those who use competitor's products
- Three categories of user (Eason, 1987):
 - **primary**: frequent hands-on
 - **secondary**: occasional or via someone else
 - **tertiary**: affected by its introduction, or will influence its purchase

Who are the stakeholders?

- Suppliers
- Local shop owners



Check-out operators



Managers and owners



Customers

What are the users' capabilities?

Humans vary in many dimensions:

- Size of hands may affect the size and positioning of input buttons
- motor abilities may affect the suitability of certain input and output devices
- height if designing a physical kiosk
- strength - a child's toy requires little strength to operate, but greater strength to change batteries
- disabilities(e.g. sight, hearing, dexterity)



What are 'needs'?

- Users rarely know what is possible
- Users can't tell you what they 'need' to help them achieve their goals
- Instead, look at existing tasks:
 - their context
 - what information do they require?
 - who collaborates to achieve the task?
 - why is the task achieved the way it is?
- Envisioned tasks:
 - can be rooted in existing behaviour
 - can be described as future scenarios



Where do alternatives come from?

- Humans stick to what they know works
- But considering alternatives is important to ‘break out of the box’
- Designers are trained to consider alternatives, software people generally are not
- How do you generate alternatives?
 - ‘Flair and creativity’: research and synthesis
 - Seek inspiration: look at similar products or look at very different products

IDEO TechBox

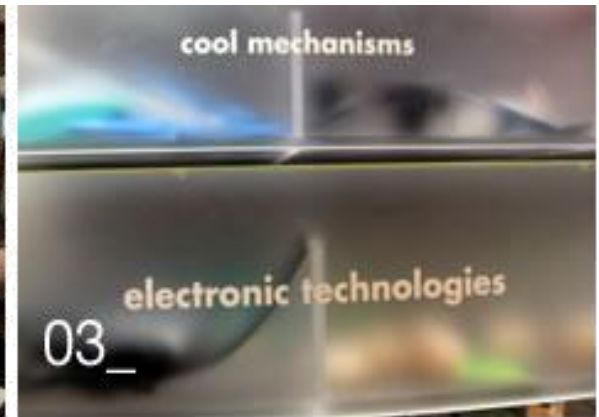
- Library, database, website - all-in-one
- Contains physical gizmos for inspiration



The Tech Box is centrally located



An item on the intranet website



The drawers are sorted by categories

From: www.ideo.com/

The TechBox



Each drawer resembles a bento box



The curator keeps order



All the entries are tagged



It really is used daily



Two demonstrations units on top

How do you choose among alternatives?

- *We choose between alternative designs by letting users and stakeholders interact with them and by discussing their experiences, preferences and suggestions for improvement*
- Evaluation with users or with peers, e.g. prototypes
- Technical feasibility: some not possible
- Quality thresholds: Usability goals lead to usability criteria set early on and check regularly
 - safety: how safe?
 - utility: which functions are superfluous?
 - effectiveness: appropriate support? task coverage, information available
 - efficiency: performance measurements

Testing prototypes to choose among alternatives



Usability engineering

- The process of writing down formal, verifiable-and hence measurable-usability criteria is a key characteristic of an approach to interaction design called usability engineering.
- Usability engineering involves specifying quantifiable measures of product performance, documenting them in a usability specification, and assessing the product against them.

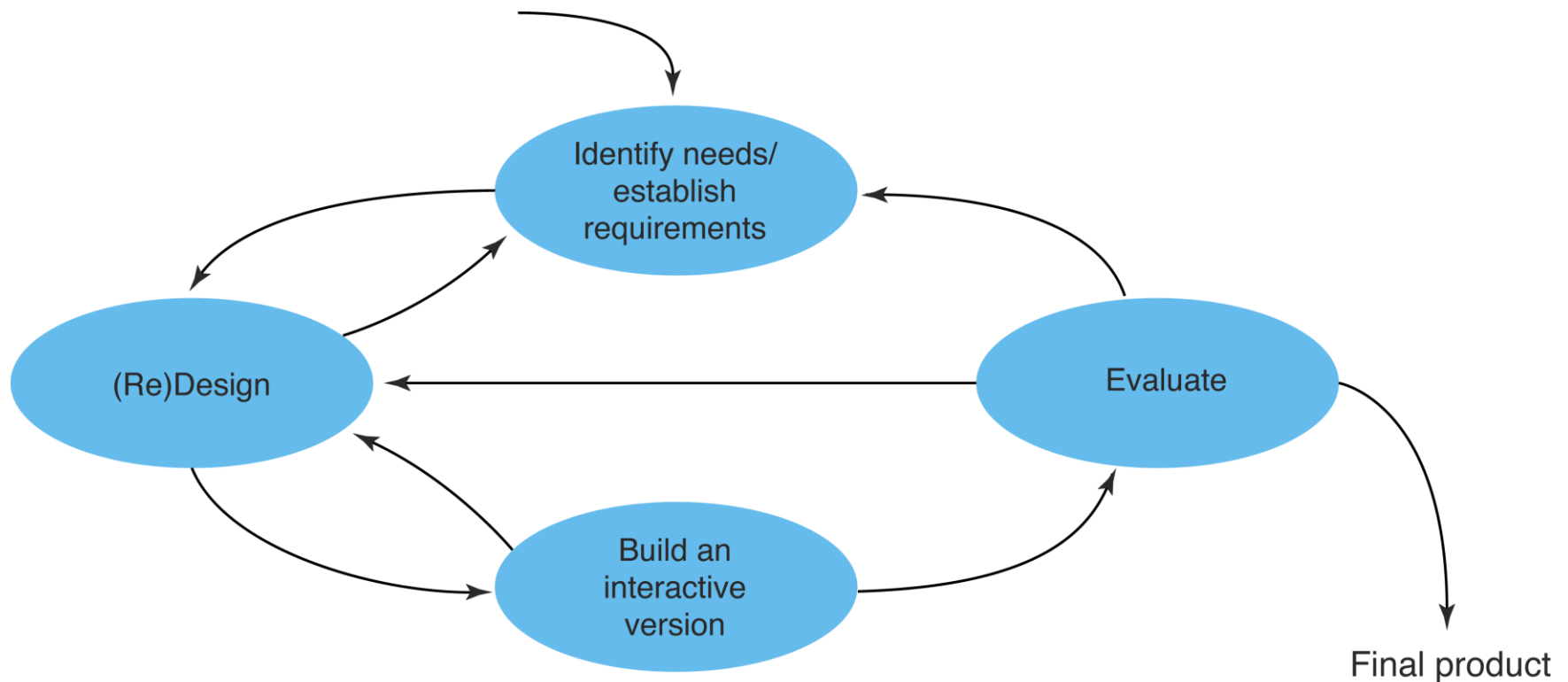
Usability criteria – quality threshold

- *Imagine that you want to design an electronic calendar or diary for yourself. You might use this system to plan your time, record meetings and appointments, mark down people's birthdays, and so on, basically the kinds of things you might do with a paper-based calendar.*
- Suggest some usability criteria that you could use to determine the calendar's quality.

Lifecycle models

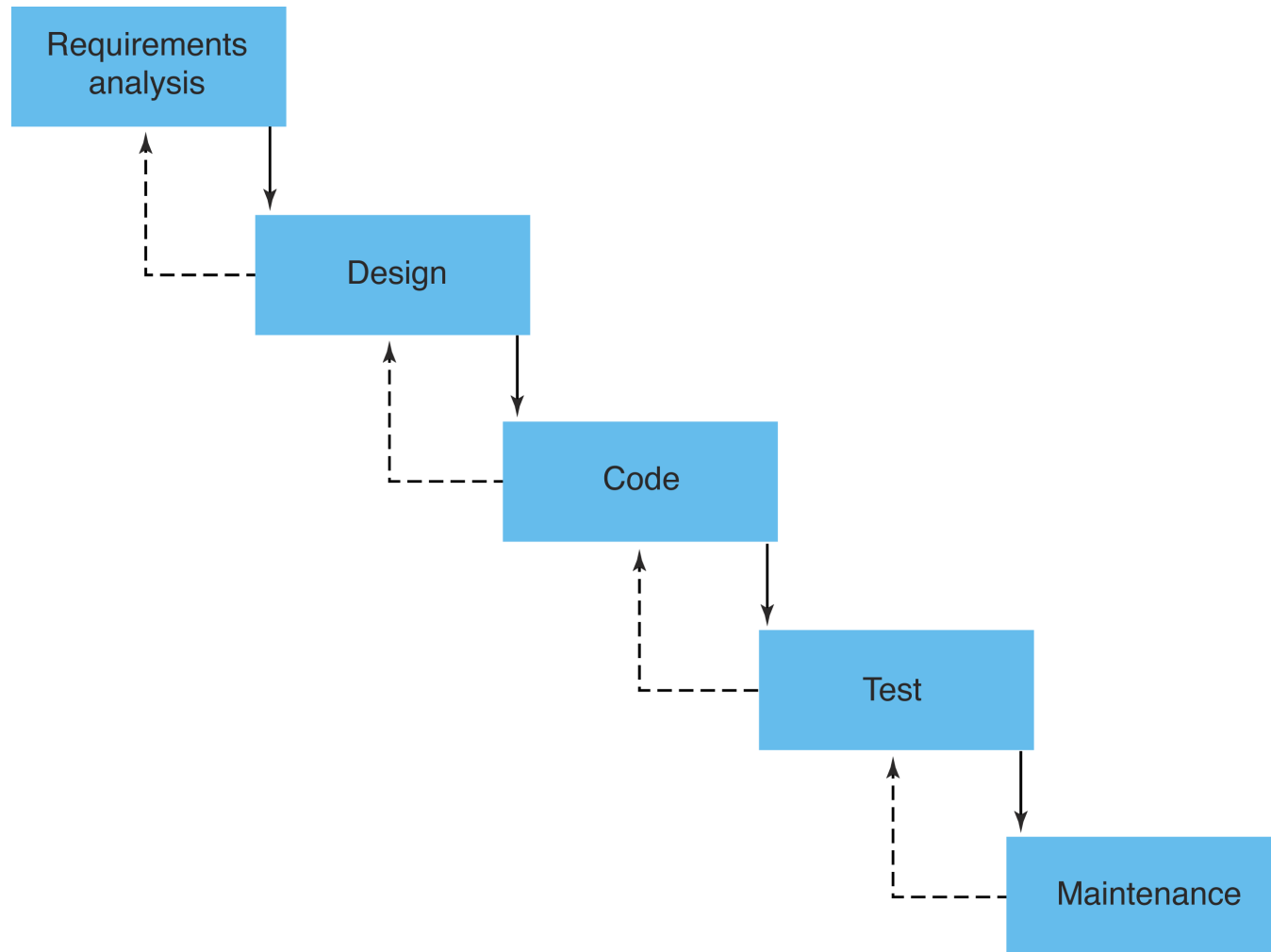
- Show how activities are related to each other
- Lifecycle models are:
 - management tools
 - simplified versions of reality
- Many lifecycle models exist, for example:
 - from software engineering: waterfall, spiral, JAD/RAD, Microsoft, agile
 - from HCI: Star, usability engineering

A simple interaction design model



Exemplifies a user-centered design approach

Traditional 'waterfall' lifecycle



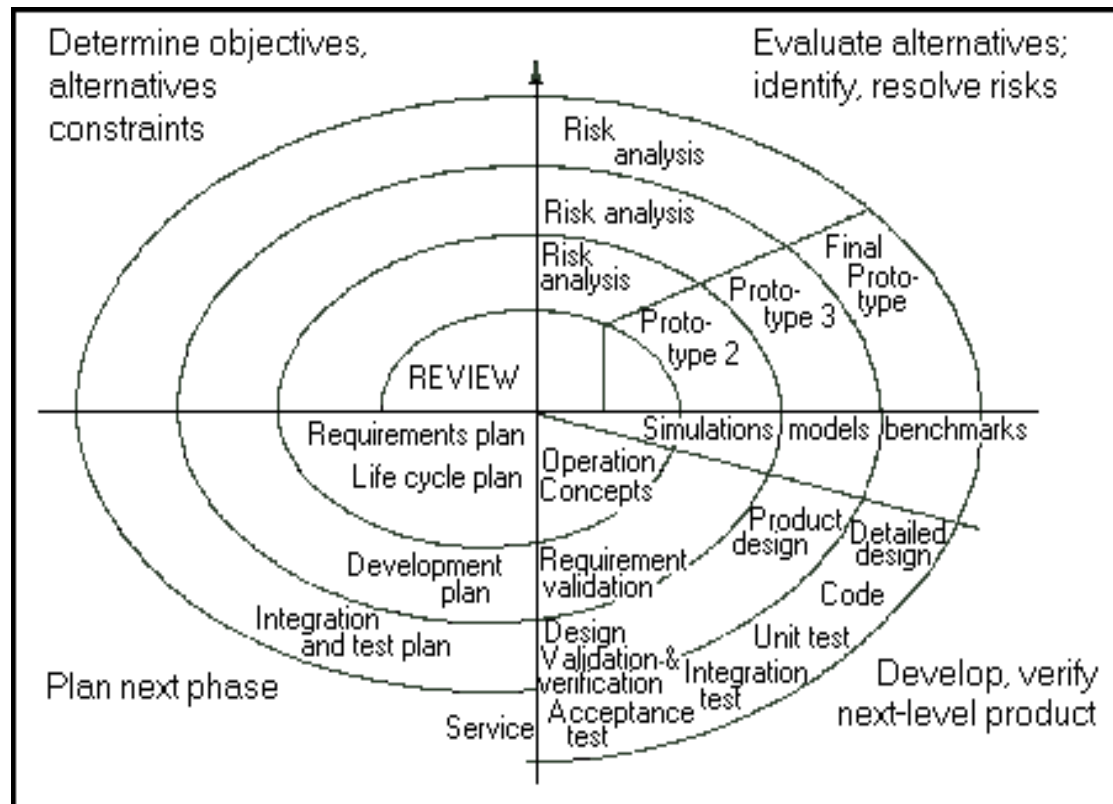
Spiral model (Barry Boehm)

Important features:

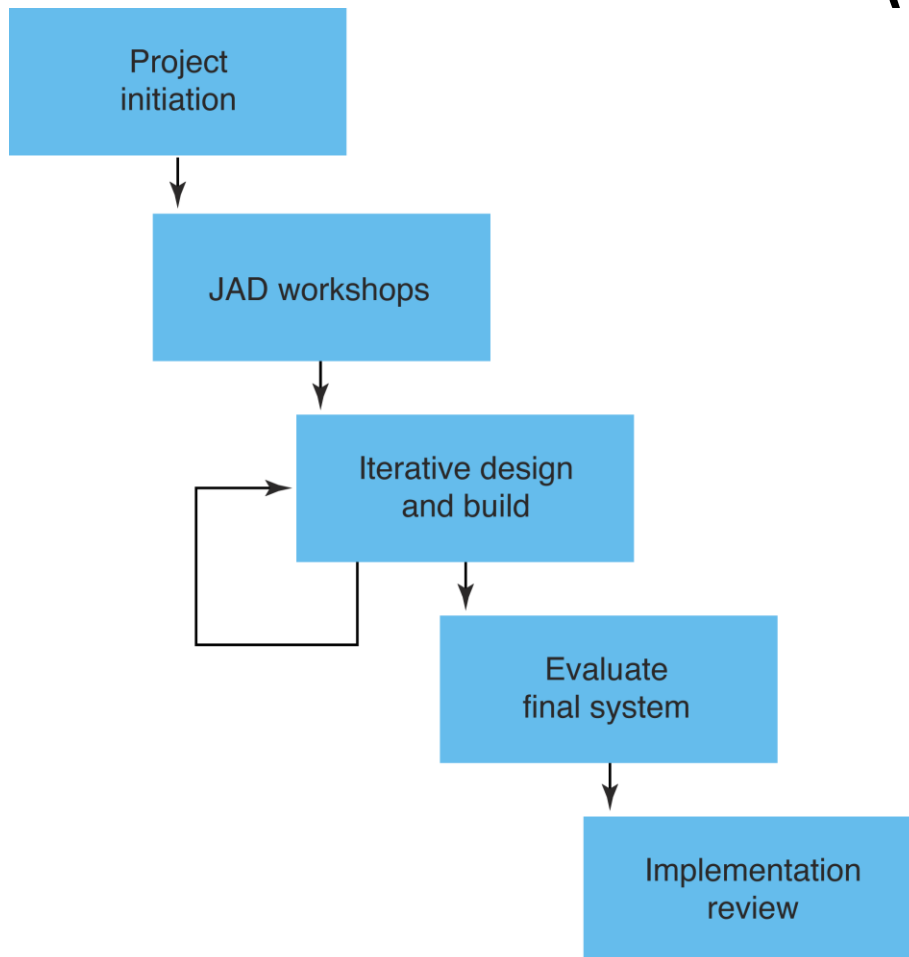
- Risk analysis
- Prototyping
- Iterative framework so ideas can be checked and evaluated
- Explicitly encourages considering alternatives

Good for large and complex projects but not simple ones

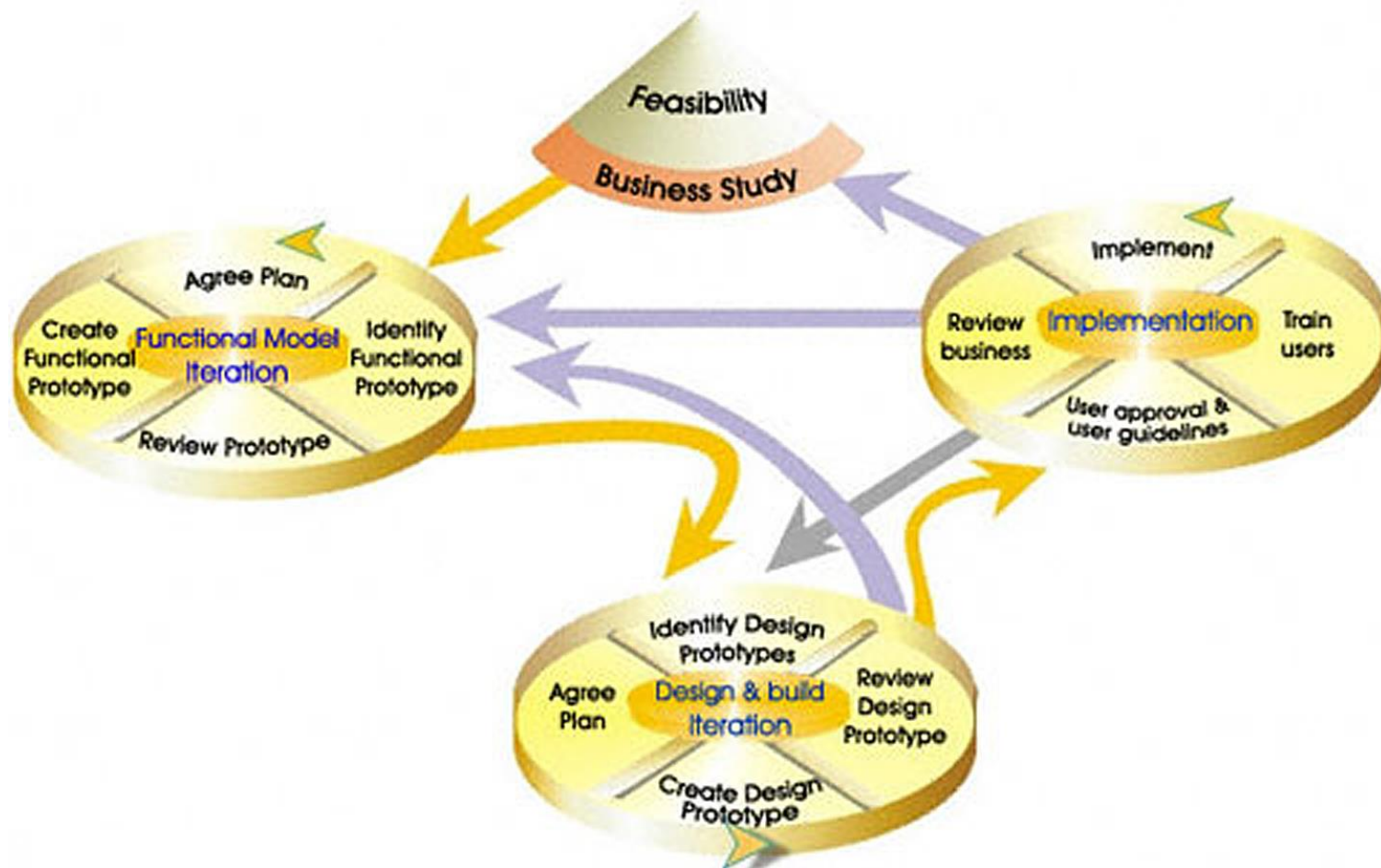
Spiral Lifecycle model



A Lifecycle for RAD (Rapid Applications Development)



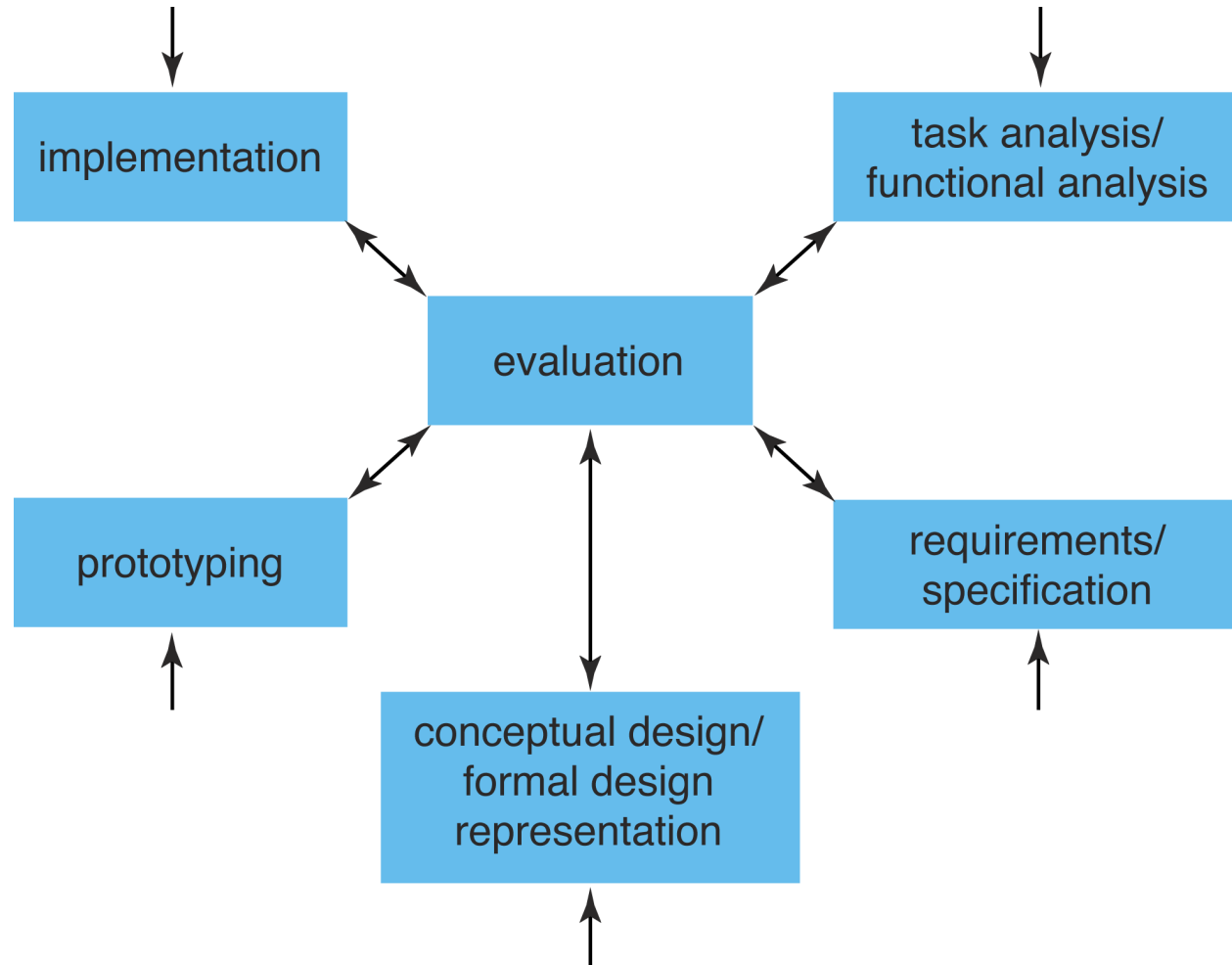
DSDM lifecycle model



The Star lifecycle model

- Suggested by Hartson and Hix (1989)
- Important features:
 - Evaluation at the center of activities
 - No particular ordering of activities; development may start in any one
 - Derived from empirical studies of interface designers

The Star Model (Hartson and Hix, 1989)



Usability engineering lifecycle model

- Reported by Deborah Mayhew
- Important features:
 - Holistic view of usability engineering
 - Provides links to software engineering approaches, e.g. OOSE
 - Stages of identifying requirements, designing, evaluating, prototyping
 - Can be scaled down for small projects
 - Uses a style guide to capture a set of usability goals

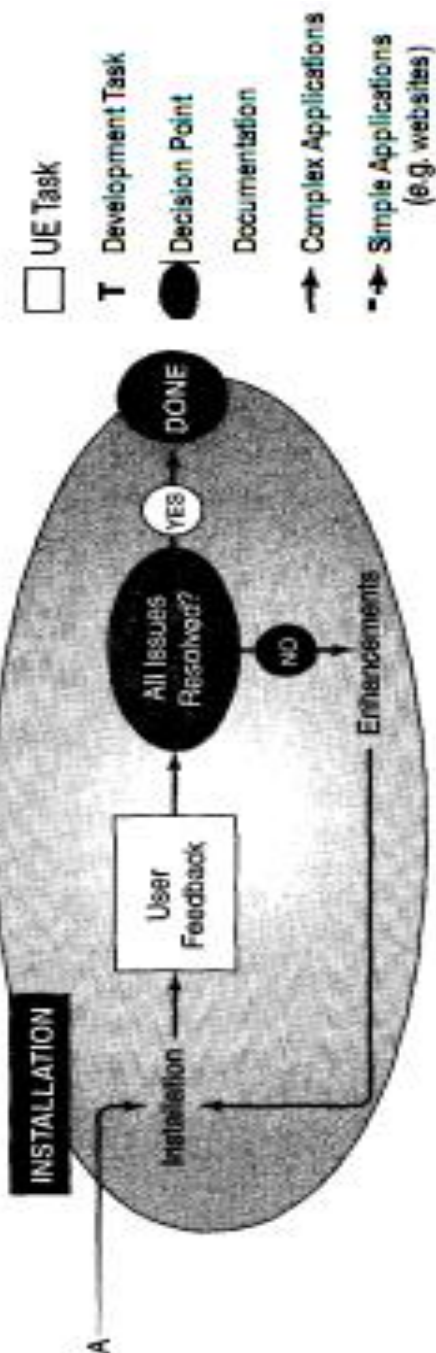
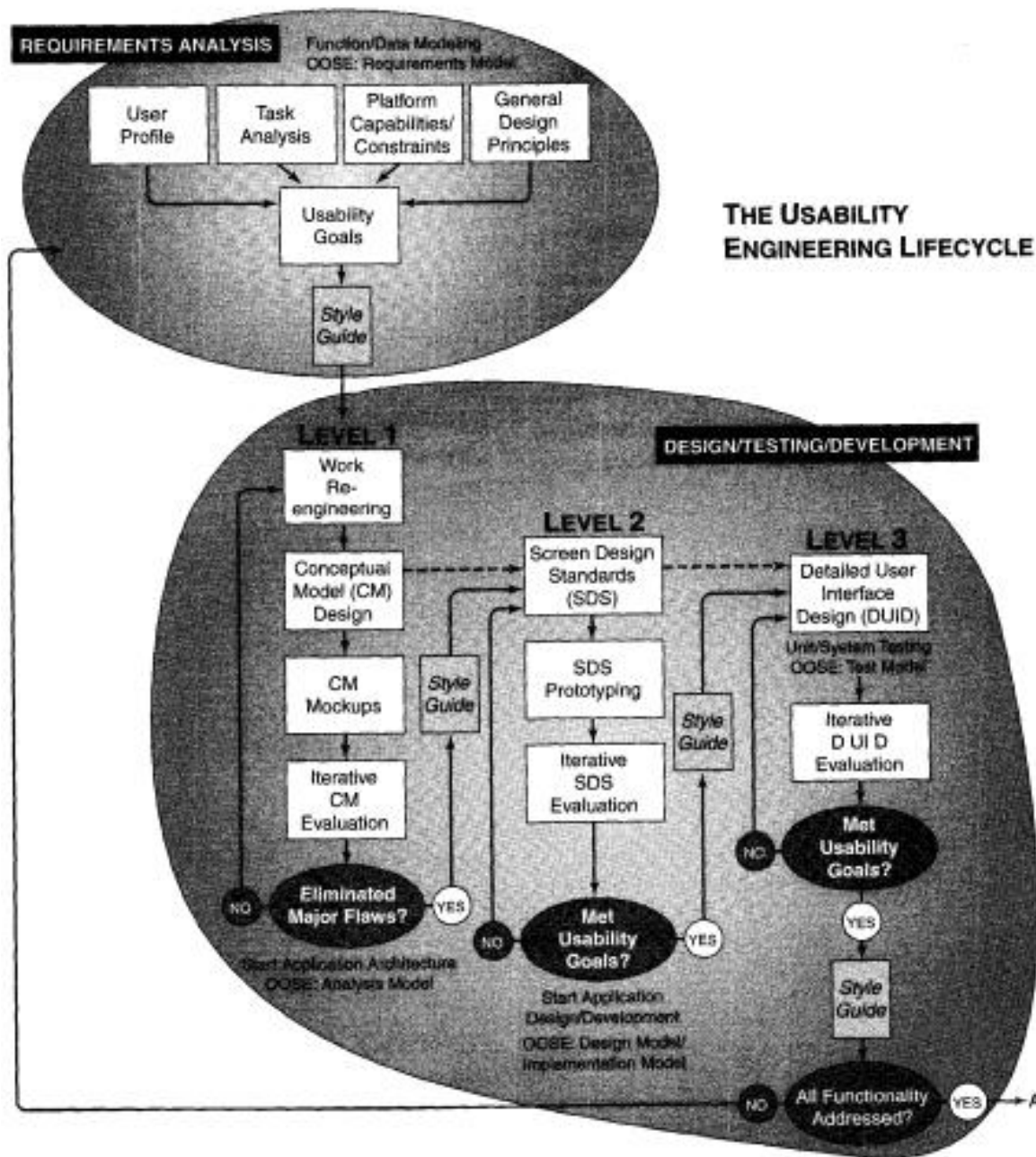
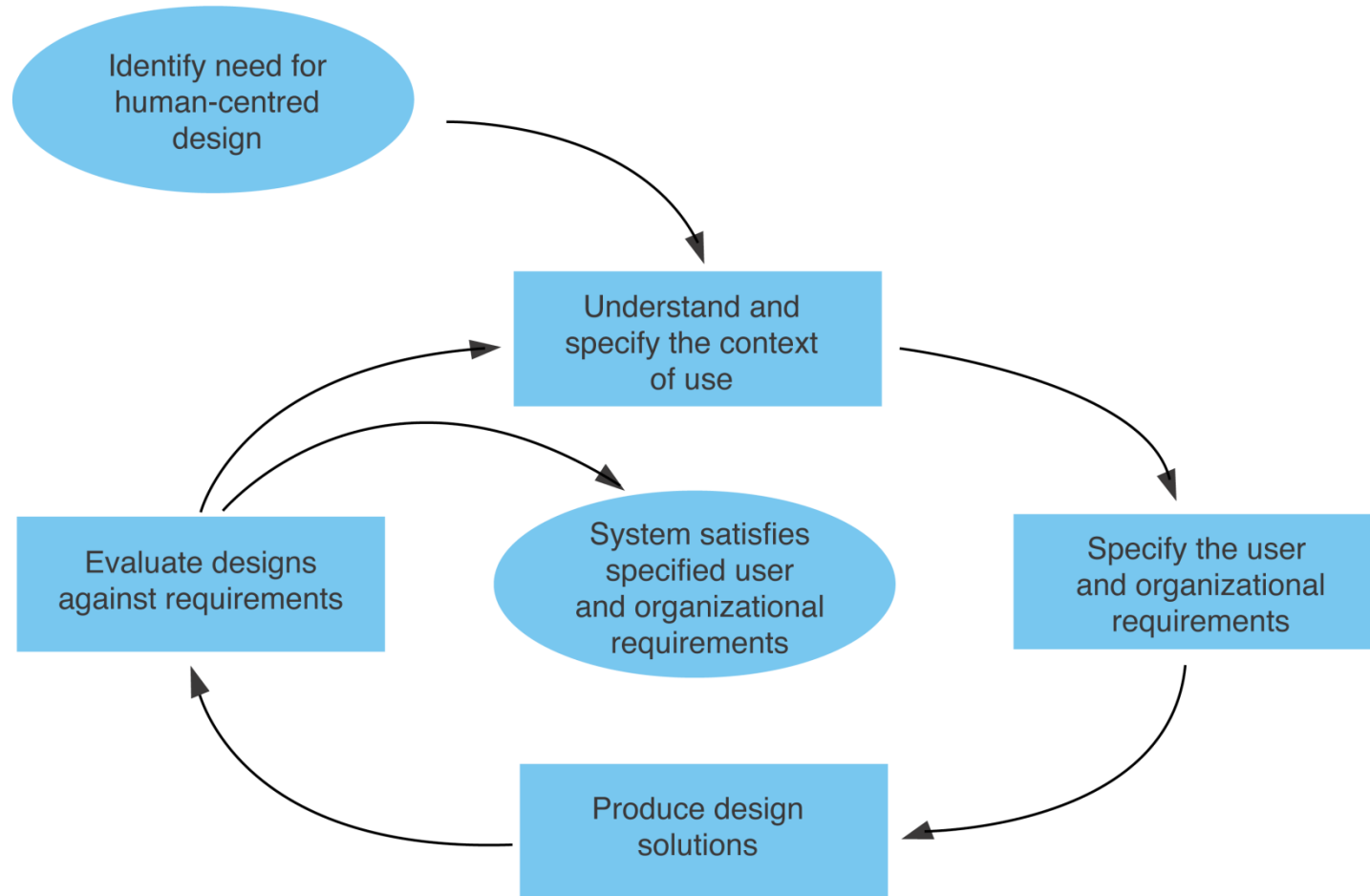


Figure 6.14 The Usability Engineering Lifecycle.

ISO 13407



Summary

Four basic activities in the design process

1. Identify needs and establish requirements
2. Design potential solutions ((re)-design)
3. Choose between alternatives (evaluate)
4. Build the artefact

User-centered design rests on three principles

1. Early focus on users and tasks
2. Empirical measurement using quantifiable & measurable usability criteria
3. Iterative design

Lifecycle models show how these are related